

Modeling Relationship between Commodity Prices and Currency Exchange Rates.

Yash Deshmukh*
Dhirubhai Ambani Institute for
Information and Communication
Technology
Gandhinagar, India
202418063@dau.ac.in

Paarth Patel†
Dhirubhai Ambani Institute for
Information and Communication
Technology
Gandhinagar, India
202418045@dau.ac.in

Prof. Amit Mankodi‡
Dhirubhai Ambani Institute for
Information and Communication
Technology
Gandhinagar, India
amit_mankodi@dau.ac.in

ABSTRACT

This project explores how commodity prices and currency exchange rates move together and whether one can help predict the other. Using historical data and econometric techniques, including stationarity checks and Granger causality tests, we aim to identify predictive relationships that can improve forecasting and understanding of global market linkages.

KEYWORDS

Commodity prices, Currency exchange rates, Granger causality, Time series analysis, Econometrics

1 INTRODUCTION

Commodity prices and currency exchange rates are closely connected in today's global economy. When the prices of key commodities like oil, gold, or agricultural products change, they can directly impact the currencies of countries that export them. For example, if oil prices rise, the Canadian dollar often strengthens because Canada is a major oil exporter. This project explores how commodity prices and currency exchange rates move together and whether one can help predict the other. Using historical data and techniques like stationarity checks and Granger causality tests, we aim to find out if changes in commodity prices can signal future changes in exchange rates, or if they simply move together at the same time. Understanding these relationships is important for investors, policymakers, and anyone who wants to better understand global financial interconnections.

2 LITERATURE REVIEW

Understanding the relationship between commodity prices and currency exchange rates has long been of interest in international finance due to its implications for pricing strategies, trade competitiveness, and risk management. Because most commodities are denominated in vehicle currencies such as the U.S. dollar, fluctuations in exchange rates can directly affect commodity prices and the terms of trade for resource-dependent economies. Prior literature emphasizes that commodity-linked currencies embed forward-looking information about global demand and supply conditions, making them potentially useful for forecasting.

Pincheira and Hardy (2013), in their study *Forecasting Aluminum Prices with Commodity Currencies*, investigated whether commodity-linked exchange rates provide predictive power for aluminum prices. Their empirical findings revealed that including exchange rate variables significantly improved forecast accuracy, particularly for longer horizons.

Chan, Tse, and Williams (2009), in *The Relationship between Commodity Prices and Currency Exchange Rates: Evidence from the Futures Markets*, explored the dynamic interactions between commodity futures prices and foreign exchange rates using cointegration analysis and Granger causality tests. They documented evidence of both short-run and long-run linkages, suggesting bidirectional causality in many cases.

Both studies underscore the interdependence between commodity and currency markets. However, gaps remain in understanding the robustness of these relationships across different commodities and during periods of heightened volatility.

3 DATASET

The research papers we studied used premium financial datasets like Thomson Reuters Datastream and Commodity Systems Inc. (CSI). Since these databases are not freely available, we had to find alternative sources. Initially, we searched for commodity datasets that included trade volumes to study both price and quantity effects. However, no reliable dataset covered all required commodities with volume data.

Eventually, we found a suitable alternative: the **World Bank's Commodity Markets Outlook (CMO) database**, which provides monthly historical prices for a wide range of commodities. This dataset is publicly available and widely used for economic research.

- **Link to dataset:** World Bank CMO Historical Data
- **Included columns:**
 - **Month/Year:** The time period for each observation.
 - **Commodity Prices:** Separate columns for each commodity (e.g. oil, gold, tea, coal, rubber), showing prices in USD per relevant unit.

For currency exchange rate data, we used historical USD-currency pair data from **Investing.com**, which included:

- **Date:** Date of each observation.
- **Price:** Exchange rate of the currency against USD.
- Other columns such as Open, High, Low, Volume, and Change %, but our analysis focused on the Price column.

*Student ID: 202418063

†Student ID: 202418045

‡Project Mentor

4 WORK DONE AND KEY FINDINGS

4.1 Data Collection and Cleaning

We collected commodity prices from the World Bank and currency exchange rates from Investing.com. After merging the datasets on date:

- We converted the commodity dataset's Month/Year format into a standard date format for proper merging.
- Missing values and non-numeric entries (e.g. ".") were converted to NaN and then to usable numeric formats.
- We ensured all columns used in analysis were numeric to avoid statistical errors.

4.2 Stationarity Checks

We performed Augmented Dickey-Fuller (ADF) tests to check for stationarity. Most series were non-stationary at their original level but became stationary after differencing, which is common in economic time series data.

4.3 Correlation Analysis

We computed Pearson correlation coefficients to check for linear relationships between commodity prices and currency exchange rates.

Key Findings:

- **Positive correlations:** South Africa's rand (ZAR) showed strong positive correlations with food commodities, matching its agricultural export profile. Iceland's krona (ISK) showed positive correlation with marine exports, which aligns with

its fisheries sector, but correlations with gold, tobacco, and banana were unexpected and need further validation. Russia's ruble (RUB) correlated positively with beef exports, fitting its trade profile.

- **Negative correlations:** Canada's dollar (CAD) usually strengthens with rising oil prices but recently has sometimes weakened despite oil price increases, possibly due to macroeconomic factors. Sorghum and platinum correlations with CAD need rechecking due to limited direct trade linkages. ISK's negative correlation with sugar prices lacked supporting literature. Russia's ruble showed typical trade-based negative correlation with sugar exports.

4.4 Granger Causality Test

We tested whether each commodity Granger-causes its corresponding currency price. Granger causality indicates predictive ability rather than true economic causation.

Next Steps:

- Verify Granger test results to ensure stability.
- Develop predictive models based on significant Granger-causal commodities to forecast currency movements.

5 CONCLUSION

Our analysis so far confirms that certain commodities and currency pairs show strong relationships, both contemporaneous and predictive. Moving forward, we plan to refine our predictive models to leverage these insights for practical forecasting applications.